



Lecture 3: Planetary Heat & Energy Transport

Planetary Systems

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Learning objectives

By the end of this lecture, you will be able to:

- Identify the main energy sources driving planetary evolution
- Distinguish conduction, convection, and radiation as heat transport mechanisms
- Derive and apply the conductive cooling timescale $\tau \sim L^2/\kappa$
- Explain the Rayleigh and Nusselt numbers and what they tell us
- Describe tidal heating and why Io is the most volcanic body in the solar system

Recap: Lecture 2

- Protoplanetary disks: structure, snow line, 3–5 Myr lifetime
- Dust → planetesimals → planets (streaming instability, pebble accretion)
- Core accretion vs. gravitational instability for giant planets
- Kepler's laws, vis-viva equation
- Orbital resonances, tidal forces, migration

Today: What happens *inside* planets after they form?

A grayscale image of the planet Io, showing its heavily cratered surface. A bright, elongated plume of material is visible on the left side of the planet, extending into the dark space. The text "1. Energy sources" is overlaid in white on the right side of the planet.

1. Energy sources

What heats a planet?

Four principal sources of internal heat:

1. **Accretional heating:** gravitational energy from assembly
2. **Gravitational differentiation:** core formation releases ~10% more
3. **Radioactive decay:** sustained for billions of years
4. **Tidal heating:** orbital energy dissipated as friction

Without internal heat → no volcanism, no magnetic field, no plate tectonics, no atmospheric outgassing.

Accretional heating

Gravitational energy of assembling a uniform sphere:

$$E_{\text{acc}} \sim \frac{3}{5} \frac{GM^2}{R}$$

For Earth: $E_{\text{acc}} \approx 2.2 \times 10^{32} \text{ J}$

If all retained as heat:

$$\Delta T = \frac{E_{\text{acc}}}{Mc_p} \approx \frac{2.2 \times 10^{32}}{5.97 \times 10^{24} \times 1200} \approx 30,000 \text{ K}$$

Enough to melt Earth many times over. Accretion inevitably produces a **magma ocean**.

Where does the accretional energy go?

Not all E_{acc} is retained as interior heat.

- **Radiation to space:** hot surface cools efficiently during late accretion
- **Thermal blanketing:** a thick primordial atmosphere traps heat, raising retention
- **Depth of deposition:** energy from giant impacts is buried deep, while small impacts deposit near the surface
- Typical retained fraction: 10–50% (depends on impact size distribution)

Even 10% retention is enough to fully melt a terrestrial planet: the magma ocean is effectively guaranteed for bodies $\geq$ Mars-sized.

Where do heat-producing elements live?

Preview for Lecture 4: elements partition differently between metal and silicate phases.

- **Lithophile** (“rock-loving”): partition into silicate phase
 - U, Th, K: the long-lived heat producers ⇒ **mantle and crust**
- **Siderophile** (“iron-loving”): partition into metallic iron
 - Fe, Ni, Co, Pt, Ir, Au ⇒ **core**; very little radiogenic heating there today

Because U, Th, K are lithophile, radiogenic heat is produced in the **mantle**, not the core, and that sustains mantle convection.

Source: Goldschmidt (1937); McDonough (2003)

Gravitational differentiation

A second, one-time burst of heat accompanies **core formation**.

- Dense iron droplets sink from the magma ocean to the centre
- Gravitational potential energy → heat via viscous friction
- Energy release ~10% of the accretional budget

Order-of-magnitude estimate for Earth:

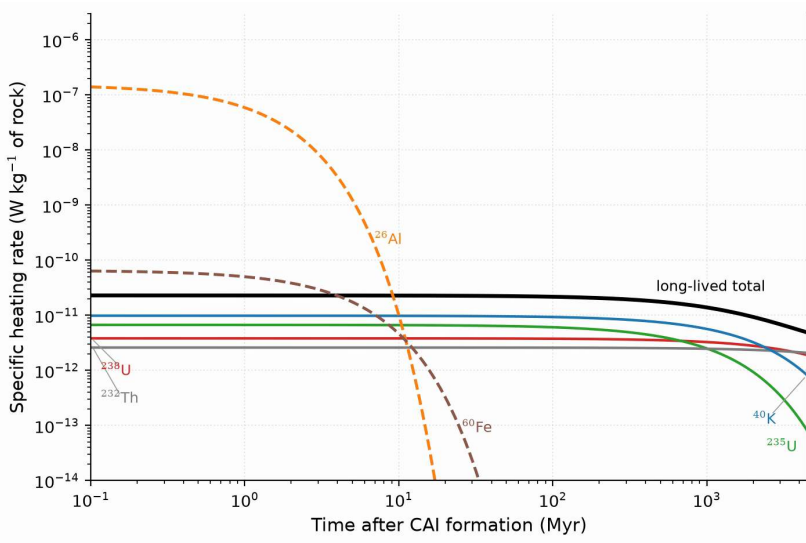
$$E_{\text{diff}} \sim \frac{3}{10} \frac{GM_{\text{core}}^2}{R_{\text{core}}} \left(\frac{R}{R_{\text{core}}} - 1 \right) \approx 2 \times 10^{31} \text{ J}$$

This heat is released **once**, during core formation, and sustains the magma ocean long enough for chemical differentiation (Lecture 4).

Radioactive decay: the long-lived engine

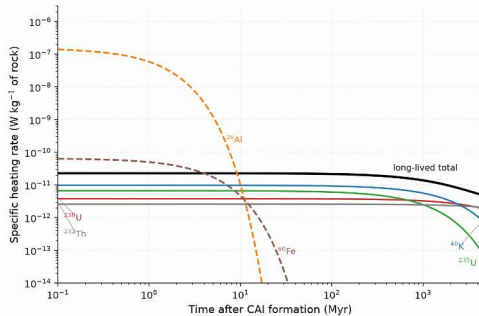
Isotope	Half-life (Gyr)	Heat (W kg^{-1})	Concentration
^{238}U	4.47	9.5×10^{-5}	20 ppb
^{235}U	0.704	5.7×10^{-4}	0.14 ppb
^{232}Th	14.0	2.6×10^{-5}	80 ppb
^{40}K	1.25	2.9×10^{-5}	28 ppb

- Lithophile elements → concentrate in mantle/crust
- Present-day radiogenic heat: ~20 TW
- 4.5 Gyr ago: almost 5× higher (factor 4.7 with these abundances)



Specific radiogenic heating from CAI formation to today, both isotope families. Course figure.

Reading the radiogenic heating curves



- **Dashed:** the short-lived pair in chondritic rock. ^{26}Al starts near $10^{-7} \text{ W kg}^{-1}$ and dies within $\sim 10 \text{ Myr}$; ^{60}Fe is three orders weaker.
- **Solid:** the four long-lived isotopes in bulk-silicate-Earth rock, summing (black) to 92 TW at formation and 19 TW today, a factor 4.7.
- The crossover near $\sim 10 \text{ Myr}$ hands the heating from the planetesimal era to the planetary era.

Two clocks drive planetary heat: a fierce Myr-scale one that melts planetesimals, and a Gyr-scale one that keeps planets alive.

Short-lived isotopes: ^{26}Al

- Half-life: only 0.72 Myr (now extinct)
- Produced by nearby stellar nucleosynthesis, delivered into the protoplanetary disk
- Initial $^{26}\text{Al}/^{27}\text{Al}$ ratio $\sim 5 \times 10^{-5}$ (canonical)
- In the first few Myr: extraordinarily potent heat source ($10^{-7} \text{ W kg}^{-1}$ initially)
- Could melt planetesimals as small as a few km radius
- Initiated core formation and volatile loss **before** planets fully assembled

^{26}Al is why even small asteroids show evidence of melting and differentiation in the meteorite record.

Where did ^{26}Al come from?

Short-lived isotopes require a nearby stellar source just before solar system formation.

- ^{26}Al is produced in:
 - Massive star winds (Wolf-Rayet stars)
 - Core-collapse supernovae
- With a 0.72 Myr half-life, it must reach the nascent solar nebula within ~1–2 Myr of production
- Implies the Sun formed in a **massive star cluster**, an Orion-like star-forming region
- ^{60}Fe : produced primarily by supernovae; detection confirms contribution from a nearby SN
- Isotopic abundances in meteorites constrain: one nearby massive star, or multiple injection events

Source: Dauphas & Chaussidon (2011); Desch et al. (2023)

Other short-lived radionuclides

The young solar system contained a full zoo of short-lived isotopes:

Isotope	Half-life (Myr)	Daughter	Role
^{26}Al	0.72	^{26}Mg	Dominant heating
^{60}Fe	2.6	^{60}Ni	Minor heating; debated initial abundance
^{53}Mn	3.7	^{53}Cr	Chronometer (not major heater)
^{182}Hf	8.9	^{182}W	Core formation chronometer (Lecture 4)
^{129}I	16	^{129}Xe	Atmosphere formation timing

The production of these isotopes likely required injection from a nearby supernova or AGB star into the molecular cloud that formed the solar system.

Extinct radionuclide chronometers

Short-lived isotopes don't just heat planetesimals; they also act as **clocks** for early differentiation.

System	Half-life	Daughter	What it dates
^{182}Hf – ^{182}W	8.9 Myr	W	Core formation
^{26}Al – ^{26}Mg	0.72 Myr	Mg	Planetesimal crystallisation
^{53}Mn – ^{53}Cr	3.7 Myr	Cr	Early differentiation events
^{129}I – ^{129}Xe	16 Myr	Xe	Atmosphere retention / volatile loss
^{244}Pu –Xe fission	81 Myr	Xe	Closure of mantle degassing

Combined, these systems pin the first ~100 Myr of solar system history to within a few Myr.

Source: Kleine & Wadhwa (2017), *Rev. Mineral. Geochem.*

How heat production has changed over time

- $t = 0$ (solar system formation): ^{26}Al dominates ($\sim 10^{-7} \text{ W kg}^{-1}$)
- $t = 5 \text{ Myr}$: ^{26}Al mostly gone
- $t = 10\text{--}100 \text{ Myr}$: transition to long-lived isotopes
- $t = 0.5 \text{ Gyr}$: ^{235}U and ^{40}K dominate
- $t = 4.5 \text{ Gyr}$ (today): ^{238}U and ^{232}Th dominate
- Total radiogenic heat today is $\sim 20\%$ of its value at formation (1/4.7)

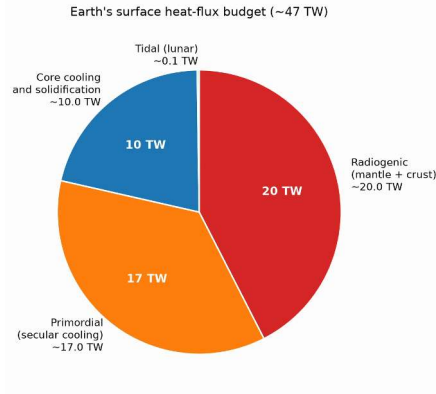
Small bodies melted early thanks to ^{26}Al , even if their long-lived budget was modest.

Earth's heat budget: 47 TW

Source	Contribution
Radiogenic heating (mantle + crust)	~20 TW
Primordial (secular cooling)	~17 TW
Core cooling and solidification	~10 TW
Tidal dissipation (from the Moon)	~0.1 TW
Total	~47 TW

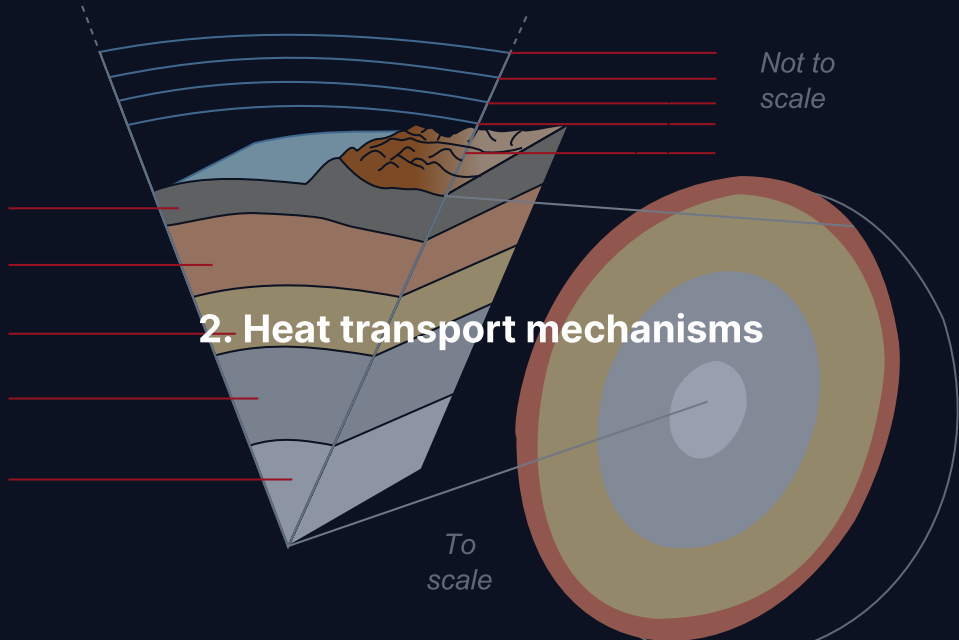
~40% from ongoing radioactive decay; 60% from primordial heat.
Earth is still cooling.

Earth's heat budget at a glance



- Radiogenic decay (red, 20 TW) and stored primordial heat (orange, 17 TW) carry most of the 47 TW.
- Core cooling and inner-core solidification add 10 TW; lunar tidal dissipation, at 0.1 TW, is invisible at this scale.
- The balance shifts over time: 4.5 Gyr ago the radiogenic slice alone was almost 5 times larger.

Earth today runs roughly 40% on radioactivity and 60% on its savings account of primordial heat.



Three ways to move heat

Conduction: Heat transferred by molecular/atomic collisions

$$\vec{q} = -k\nabla T \text{ (Fourier's law)}$$

Slow, diffusive; dominant in lithosphere, small bodies

Convection: Heat transported by bulk fluid motion

Hot material rises, cool material sinks

Dominant in mantles, outer cores, giant planet interiors

Radiation: Energy carried by electromagnetic waves

$$F = \epsilon\sigma T^4 \text{ (Stefan-Boltzmann)}$$

Dominant at surfaces and in atmospheres

Conduction: Fourier's law

- Heat flux: $\vec{q} = -k\nabla T$
- Thermal diffusivity: $\kappa = k/(\rho c_p) \approx 10^{-6} \text{ m}^2 \text{ s}^{-1}$ (rock)
- Inherently **diffusive**, slow over large distances
- The time for heat to conduct through distance L :

$$\tau_{\text{cond}} \sim \frac{L^2}{\kappa}$$

We will derive this next.

Thermal conductivity: typical values

Material	k ($\text{W m}^{-1} \text{K}^{-1}$)	κ ($\text{m}^2 \text{s}^{-1}$)
Silicate rock (upper mantle)	3–5	$\sim 10^{-6}$
Bridgmanite (lower mantle)	5–10	$\sim 2 \times 10^{-6}$
Water ice	2.2	1.1×10^{-6}
Iron (solid)	80	$\sim 2 \times 10^{-5}$
Air (atmosphere)	0.025	$\sim 2 \times 10^{-5}$

- Metals have high k but also high ρc_p : intermediate κ
- Rocks have low k and low ρc_p : similar κ
- This is why the L^2/κ scaling matters more than k alone

When does radiation dominate?

The Stefan–Boltzmann law $F = \epsilon \sigma T^4$ gives the heat flux from a surface at temperature T .

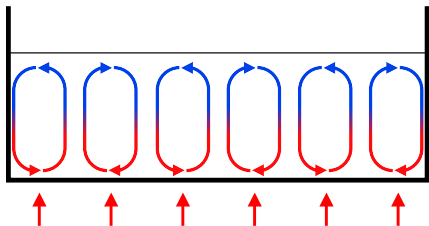
Radiation matters in:

- **Planetary surfaces:** heat loss to space (Lecture 5 energy balance)
- **Optically thin atmospheres:** e.g. Mars' upper atmosphere
- **Magma oceans:** molten silicate at $T \gtrsim 2000$ K radiates efficiently

Why not in solid interiors?

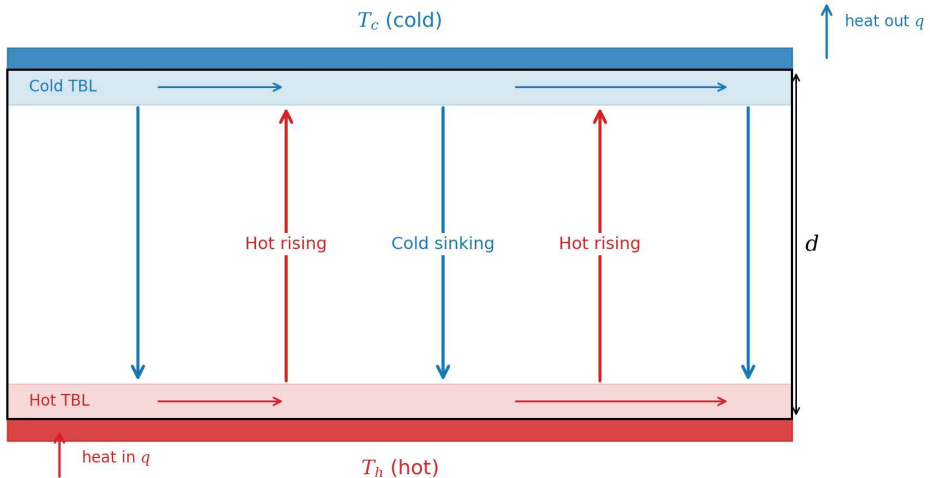
- Silicate rocks are opaque at infrared wavelengths
- Photon mean free paths are much smaller than conductive length scales
- Exception: partially transparent silicates above ~ 2000 K become radiatively conductive

Convection: nature's heat elevator



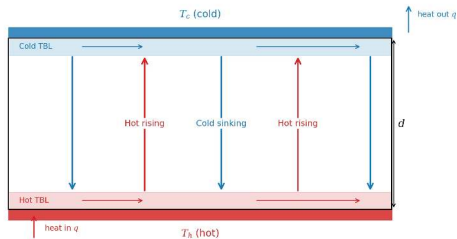
Credit: Wikimedia Commons, CC BY-SA 3.0

- Hot material at base: buoyant, rises
- Cool material at top: dense, sinks
- Far more efficient than conduction
- Active in:
 - Earth's mantle (~cm/yr)
 - Outer core (liquid metal)
 - Giant planet interiors
 - Atmospheres



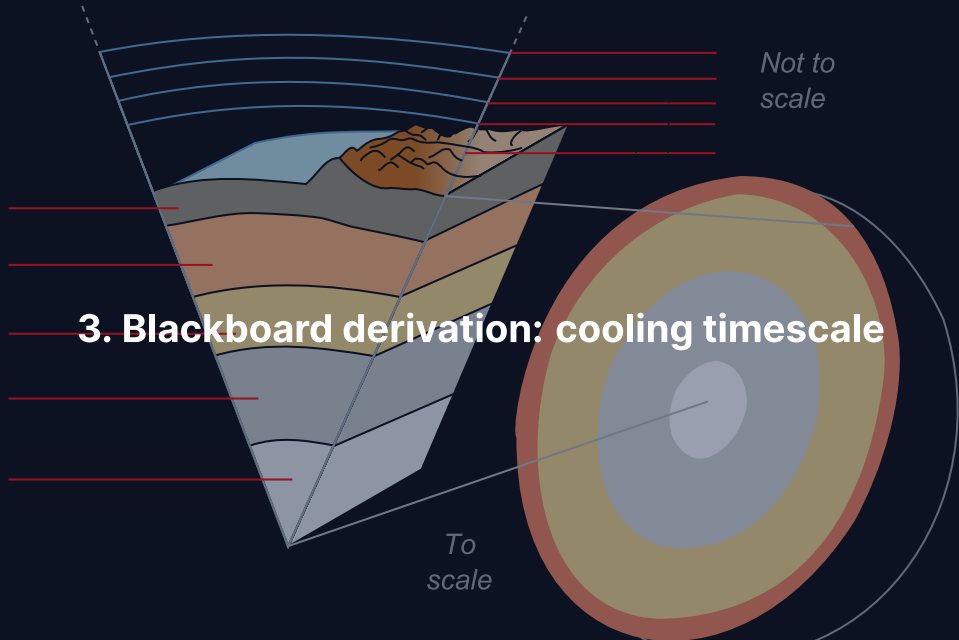
Rayleigh-Bénard convection: a fluid layer heated from below, cooled from above. Course schematic.

Reading the convection schematic



- Hot plumes rise from a thin hot **thermal boundary layer** at the base; cold fluid sinks from a thin cold one at the top.
- Lateral flow along the boundary layers closes the circulation cells.
- Convection switches on once the Rayleigh number exceeds the critical value Ra_c .

The whole story of mantle convection is in this sketch: thin conductive skins, a well-mixed interior, and overturning cells between them.



Goal and strategy

Goal: Estimate how long it takes heat to conduct out of a body of size L .

Strategy: Start from the heat equation, then use dimensional analysis instead of solving the PDE.

- We want a characteristic timescale $\tau_{\text{cond}}(L, \kappa)$
- Plug in representative values for rock
- Compare to the age of the solar system (4.6 Gyr)

Physical intuition: bigger bodies have a longer path for heat to diffuse, so they stay hot for longer. We will derive exactly *how much* longer.

Step 1: the heat diffusion equation

Fourier's law $\vec{q} = -k\nabla T$ combined with energy conservation gives:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T)$$

For constant thermal conductivity k , define $\kappa = k/(\rho c_p)$:

$$\frac{\partial T}{\partial t} = \kappa \nabla^2 T$$

In one dimension:

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2}$$

The rate of temperature change is proportional to the **curvature** of the temperature profile; flat profiles do not evolve, curved ones relax.

Step 2: dimensional analysis

Replace derivatives with characteristic scales:

- Time derivative: $\frac{\partial T}{\partial t} \sim \frac{T}{\tau}$
- Spatial derivative: $\frac{\partial^2 T}{\partial x^2} \sim \frac{T}{L^2}$

Substitute into the heat equation:

$$\frac{T}{\tau} \sim \kappa \frac{T}{L^2}$$

Temperature cancels: the timescale is intrinsic, not tied to any particular T .

$$\tau_{\text{cond}} \sim \frac{L^2}{\kappa}$$

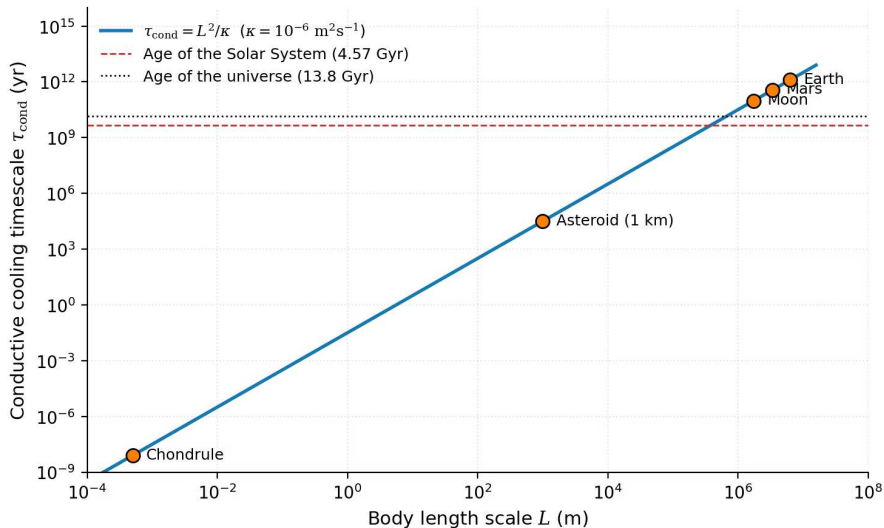
Key feature: the L^2 scaling. Doubling the body size $\rightarrow$ 4 $\times$ the cooling time.

Cooling timescales across the solar system

Using $\kappa \approx 10^{-6} \text{ m}^2 \text{ s}^{-1}$ (silicate rock):

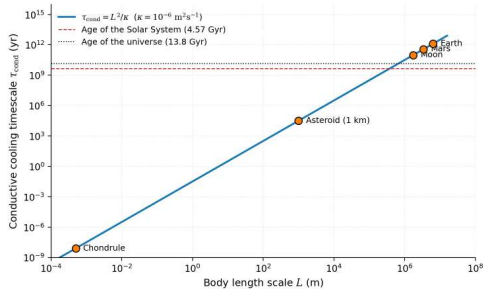
Body	Size L	$\tau = L^2 / \kappa$	Familiar units
1 m boulder	1 m	10^6 s	~12 days
100 km asteroid	10^5 m	10^{16} s	~300 Myr
Moon	$1.7 \times 10^6 \text{ m}$	$3 \times 10^{18} \text{ s}$	~100 Gyr
Earth	$6.4 \times 10^6 \text{ m}$	$4 \times 10^{19} \text{ s}$	~1300 Gyr

Earth's conductive cooling time $\gg$ age of the universe.
Yet Earth loses 47 TW $\Rightarrow$ there **must** be convection.



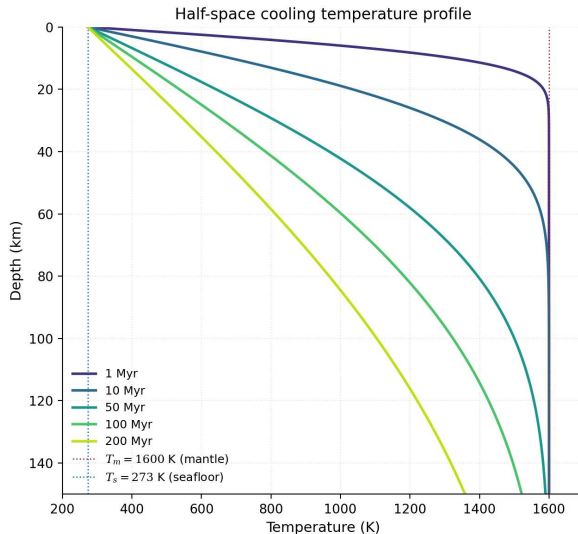
The conductive cooling timescale against body size, from chondrules to Earth. Course figure.

Reading the cooling-timescale plot



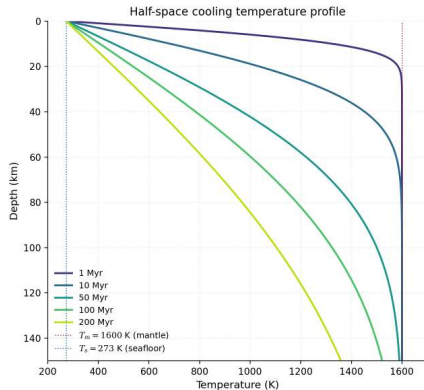
- The line is $\tau = L^2 / \kappa$ on log-log axes: the L^2 scaling appears as slope 2.
- Horizontal dashes mark the ages of the solar system (4.57 Gyr) and the universe (13.8 Gyr).
- A chondrule (0.5 mm) cools in under a second; a 1 km asteroid needs ~30 000 yr; the Moon, Mars, and Earth plot far above both age lines.

Any body larger than a few hundred km cannot have cooled by conduction: it must convect, or stay hot.



The half-space cooling model: conductive cooling of oceanic lithosphere. Course figure, after Turcotte & Schubert (2002).

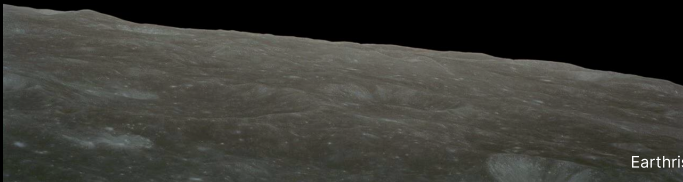
Reading the half-space cooling profiles

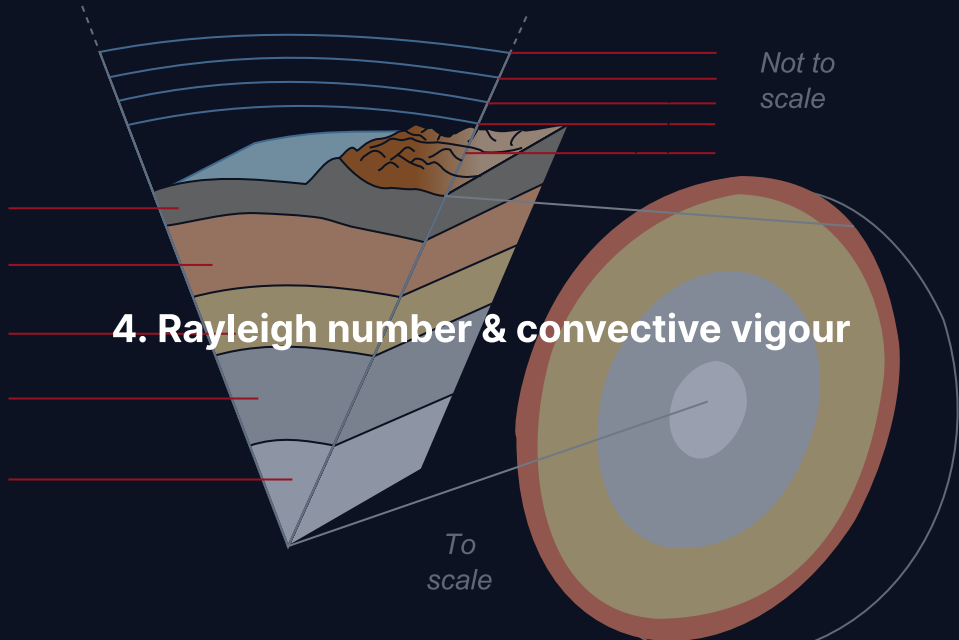


- Each curve is $T(z, t) = T_s + (T_m - T_s) \operatorname{erf}(z/2\sqrt{kt})$ for seafloor ages 1 to 200 Myr.
- The cold boundary layer (the **lithosphere**) thickens as $\sqrt{kt}$: fast when young, slower as it ages.
- The same solution predicts seafloor heat flux $q \propto t^{-1/2}$ and, through thermal contraction, ocean depth increasing with age.

One erf-solution of the heat equation explains lithosphere thickness, seafloor heat flow, and ocean bathymetry at once.

Break





The Rayleigh number

Does a fluid layer convect or merely conduct?

$$\text{Ra} = \frac{\alpha \rho g \Delta T d^3}{\kappa \eta}$$

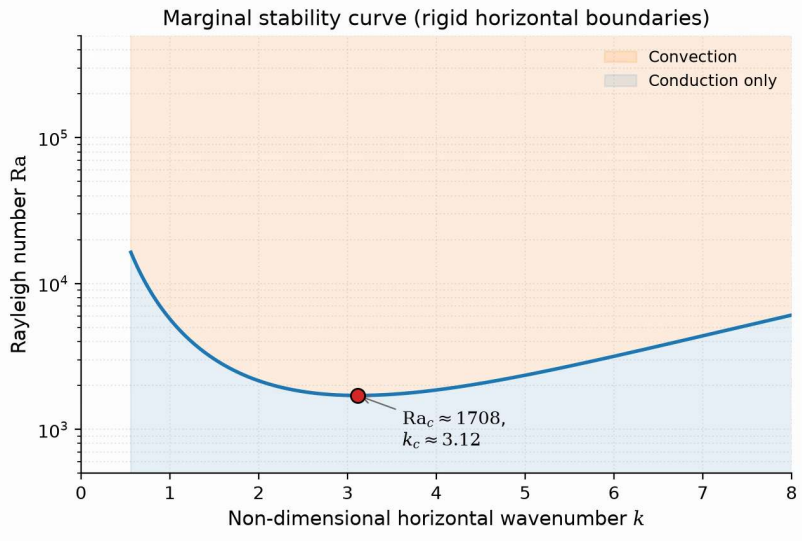
Numerator (buoyancy):

- α = thermal expansion
- ρ = density
- g = gravity
- ΔT = temperature contrast
- d = layer depth

Denominator (dissipation):

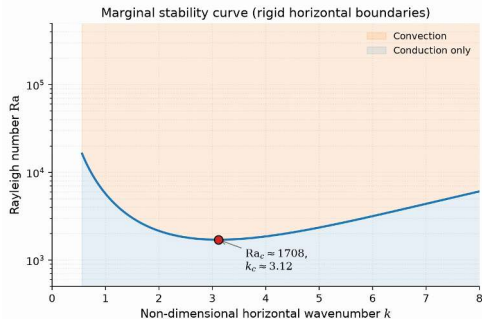
- κ = thermal diffusivity
- η = dynamic viscosity

Critical value: $\text{Ra}_c \approx 1708$
(rigid boundaries; 658 for free-slip)



The marginal stability curve for Rayleigh-Bénard convection. Course figure, after Turcotte & Schubert (2002).

Reading the marginal stability curve



- Below the curve, every perturbation decays: conduction is the only steady state.
- Above it, convection grows; the **minimum** of the curve is $Ra_c \approx 1708$, reached at wavenumber $k_c \approx 3.12$.
- The curve rises on both sides: very wide and very narrow cells are both harder to drive than the preferred cell size.

Convection is a threshold phenomenon with a preferred wavelength, set by the layer depth.

Earth's mantle: vigorous convection

For Earth's mantle:

$$Ra_{\oplus} \sim \frac{2 \times 10^{-5} \times 4000 \times 10 \times 2500 \times (3 \times 10^6)^3}{10^{-6} \times 10^{21}} \sim 10^7 - 10^8$$

- $Ra_{\oplus} \gg Ra_c \approx 1708$: **vigorous convection confirmed**
- Explains why Earth's interior cools efficiently despite conduction being inadequate

The Nusselt number $Nu = (Ra/Ra_c)^{1/3}$ measures convection efficiency. For Earth: $Nu \approx 18-39$, so convection moves heat tens of times faster than conduction.

Rayleigh number across the solar system

Body	Ra	Regime
Earth's mantle	10^7 – 10^8	Vigorous convection
Venus' mantle	10^7 – 10^8	Vigorous (stagnant lid)
Mars' mantle	10^6 – 10^7	Moderate convection
Moon's mantle	10^4 – 10^5	Weak; cooling by conduction
Mercury's mantle	$\sim 10^4$	Marginal; largely cooled
Magma ocean	$\geq 10^{25}$	Turbulent convection
Earth's outer core	$\geq 10^{25}$	Turbulent (dynamo)

Ra is hyper-sensitive to size ($\propto d^3$) and viscosity ($\propto 1/\eta$). Small bodies and very viscous interiors quickly drop below Ra_c .

Nusselt number: efficiency of convection

- $Nu = q_{\text{total}}/q_{\text{cond}}$ = actual flux over conductive flux
- $Nu = 1$: pure conduction
- $Nu \gg 1$: vigorous convection

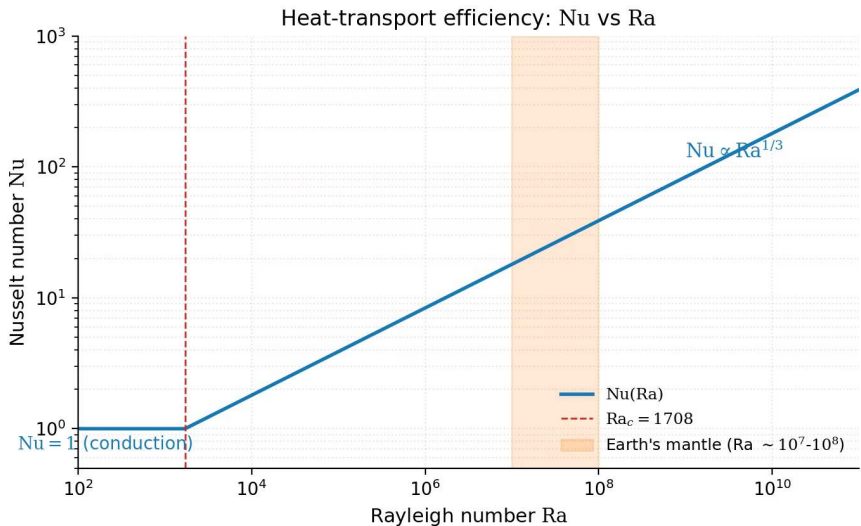
Classical scaling from boundary-layer theory:

$$Nu \propto Ra^{\beta}, \quad \beta \approx 1/3$$

Consequences:

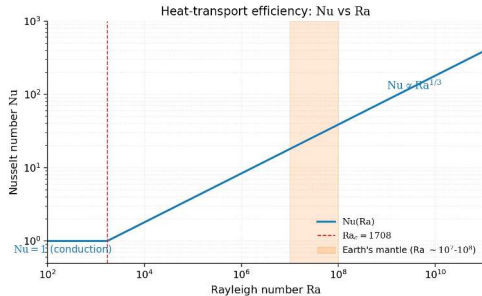
- Doubling Ra only ~26% more efficient (weak sensitivity)
- Increasing Ra by 10^3 : Nu increases by $10\times$
- Earth, with the scaling normalised to vanish at Ra_c :
 $Nu = (Ra/Ra_c)^{1/3} \approx 18\text{--}39$

Source: Schubert et al. (2001), *Mantle Convection in the Earth and Planets*



Heat-transport efficiency: the Nusselt number against the Rayleigh number. Course figure.

Reading the Nu-Ra diagram



- Below Ra_c (dashed red): $Nu = 1$, conduction only.
- Above it, the boundary-layer scaling $Nu \propto Ra^{1/3}$ takes over as a straight line on log-log axes.
- The shaded band marks Earth's mantle at $Ra \sim 10^7$ – 10^8 : 18 to 39 times more efficient than conduction.

Convection outruns conduction by a factor of tens, but the weak $1/3$ power means efficiency grows only gently with driving: even vigorous mantles need Gyr to cool.

Thermal boundary layers

A convecting interior develops three layers:

1. **Cold boundary layer (top):** conductive, steep T drop
→ Earth's **lithosphere** (~100–200 km)
2. **Well-mixed interior:** nearly adiabatic, slow T variation
→ bulk of the mantle
3. **Hot boundary layer (bottom):** steep T rise into the core
→ Earth's **D" layer** (~200 km)

Nearly all the temperature drop occurs in the two thin boundary layers.

The adiabatic gradient

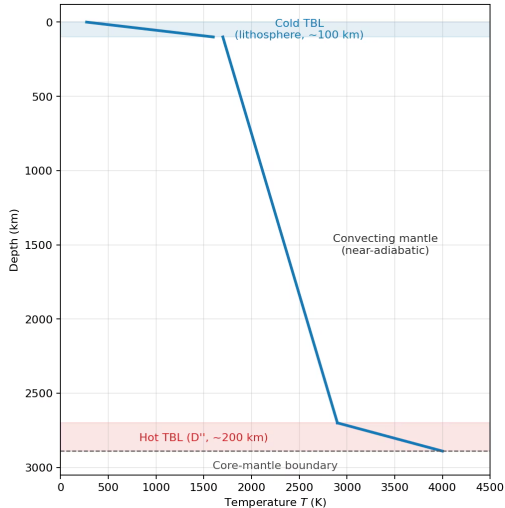
In the well-mixed convecting interior, temperature follows an **adiabat**:

$$\left(\frac{dT}{dr}\right)_{\text{adiabat}} = -\frac{\alpha g T}{c_p}$$

- α = thermal expansion, g = gravity, c_p = specific heat
- For Earth's upper mantle: ~0.3 K/km (a few K per km)
- Much shallower than the conductive gradient in the lithosphere (~25 K/km)
- Adiabatic profiles are nearly isothermal on the scale of the mantle

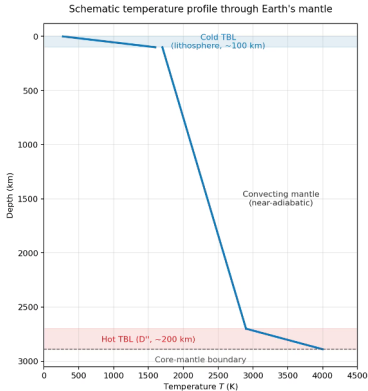
Temperature changes only a few hundred K over the entire convecting mantle: efficient mixing keeps the interior nearly adiabatic.

Schematic temperature profile through Earth's mantle



Temperature through Earth's mantle: two thin boundary layers, one adiabatic interior.
Schematic after Schubert et al. (2001).

Reading the mantle temperature profile



- The cold boundary layer (lithosphere, ~100 km) carries a steep conductive gradient from the surface into the mantle.
- The bulk mantle follows a near-adiabatic $\sim 0.5 \text{ K km}^{-1}$: nearly isothermal on this scale.
- The hot D'' layer (~200 km) bridges the jump from the mantle adiabat (~2500 K) to the core-mantle boundary (~4000 K).

Almost the entire mantle temperature drop lives in two thin skins; the interior in between is convectively flattened.

Two tectonic regimes

Mobile lid (plate tectonics):

- Cold boundary layer **breaks** into plates
- Plates subduct, recycle material
- Most efficient heat loss
- **Only Earth**

Stagnant lid:

- Single thick, immobile shell
- Heat escapes by conduction + volcanism
- Less efficient cooling
- Venus, Mars, Mercury, Moon

Why Earth is unique: likely a combination of water, size, and thermal history.

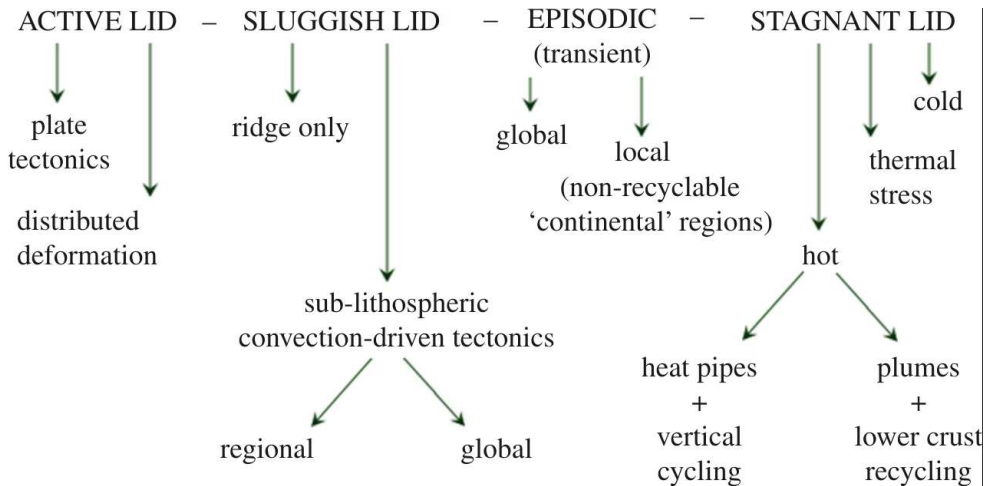


Figure 2. Potential tectonic modes for terrestrial planets.

The modern taxonomy of tectonic modes. Reproduced from Lenardic (2018), Fig. 2.

Reading the tectonic-mode taxonomy

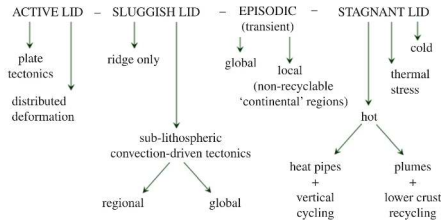
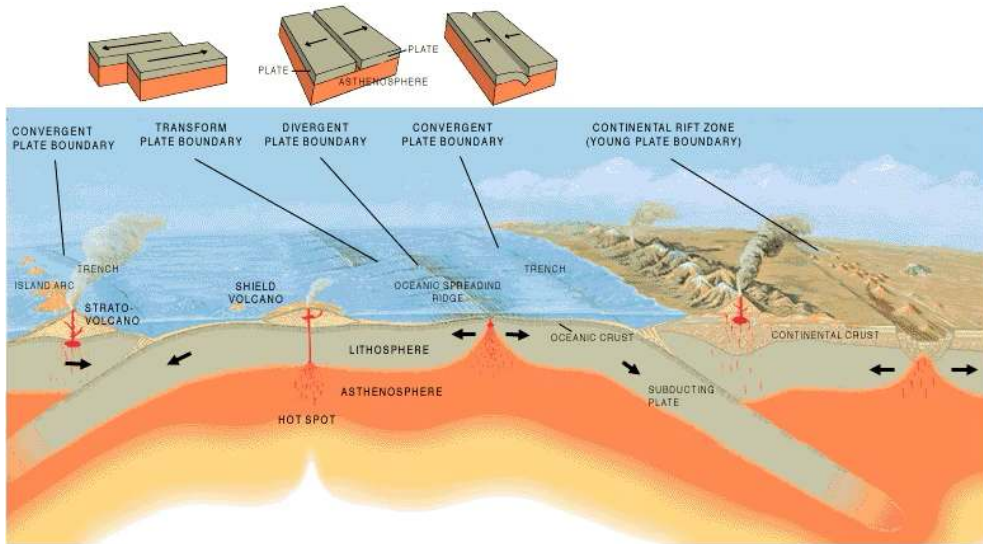


Figure 2. Potential tectonic modes for terrestrial planets.

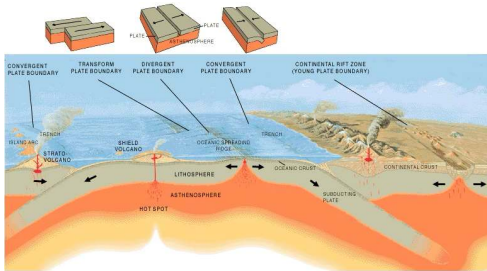
- **Active lid** splits into plate tectonics, ridge-only, and distributed deformation.
- **Episodic** modes are transient, global or local; **stagnant lids** come in cold (thermal-stress) and hot (heat-pipe, plume-fed) variants.
- The mobile-vs-stagnant binary of the previous slide is the two end-members of this richer spectrum.

Tectonic regime is a spectrum, and a planet can move through it as it cools.



The plate-tectonic settings in one cross-section. Credit: USGS, public domain.

Reading the plate-boundary cross-section



- Divergent ridges: mantle upwells, new oceanic crust forms.
- Convergent margins: the old, cold plate subducts, feeding arc volcanism above the sinking slab.
- A hot-spot plume rises from depth independently of the plate pattern.
- The lithosphere in this picture *is* the cold thermal boundary layer of mantle convection.

Plate tectonics is mantle convection seen from above: subduction is the cold downwelling.

Why does Earth have plate tectonics?

One of the biggest open questions in planetary geophysics.

Proposed factors:

- **Water:** weakens rocks at fault zones; lowers melting point in subducting slabs
- **Planet size:** higher convective stresses in a larger mantle
- **Thermal state:** must be cold enough for a rigid lid but hot enough to sustain plume-driven rifting
- **Prior history:** once plate tectonics starts, it may be self-sustaining (recycling weakens the lid)
- Venus may have had plate tectonics in the past (tessera highlands); lost it after drying out

Habitability-oriented view: plate tectonics is linked to liquid water, and both are linked to long-term climate stability.

Heat-pipe volcanism

An alternative heat-loss regime for very hot bodies.

- Magma rises through conduits (“heat pipes”)
- Erupts on the surface, cools, and is buried by new lava
- Old crust subsides into the hot interior as new lava covers it
- **lo today:** global resurfacing by volcanism every $\sim 10^6$ yr
- **Early Earth:** may have been a heat-pipe regime (Hadean, >4 Ga)
- Allows efficient cooling of a very hot mantle without plate tectonics

Source: Turcotte (1989); Moore & Webb (2013), *Nature*

Mantle plumes and hotspots

Whole-mantle convection is dominated by sheet-like downwellings (subducting slabs), but the hot bottom boundary layer is also unstable.

- Hot, buoyant material collects at the base of the mantle (D'')
- Periodically detaches as narrow, rising **plumes** (~100 km diameter)
- Reach the lithosphere and generate **intraplate volcanism**
- Plumes are long-lived (10^7 – 10^8 yr) and approximately fixed in the deep mantle

Examples (Earth): Hawaii, Iceland, Yellowstone, Réunion.

Plumes move heat from the core–mantle boundary to the surface independently of plate tectonics.

EXPLANATION

-  **Divergent plate boundaries—**
Where new crust is generated
as the plates pull away from
each other.
-  **Convergent plate boundaries—**
Where crust is consumed in the
Earth's interior as one plate
dives under another.
-  **Transform plate boundaries—**
Where crust is neither produced
nor destroyed as plates slide
horizontally past each other.
-  **Plate boundary zones—**Broad
belts in which deformation is
diffuse and boundaries are not
well defined.
-  **Selected prominent hotspots**

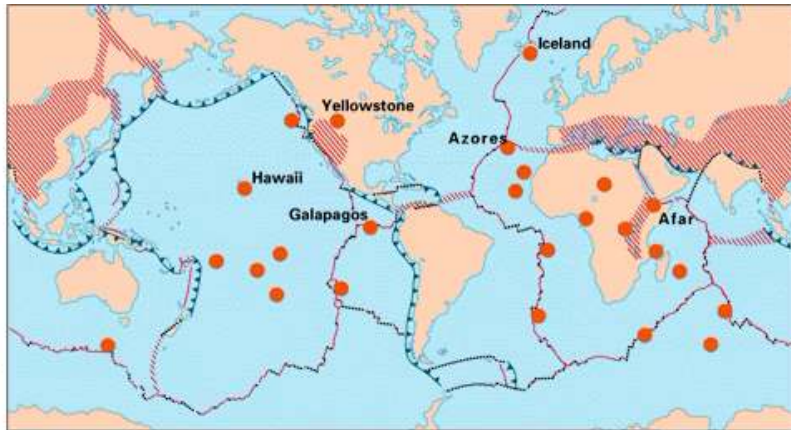
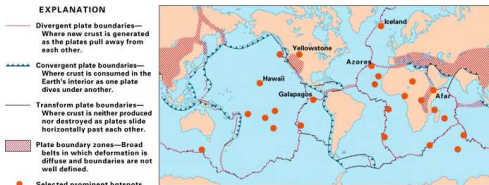


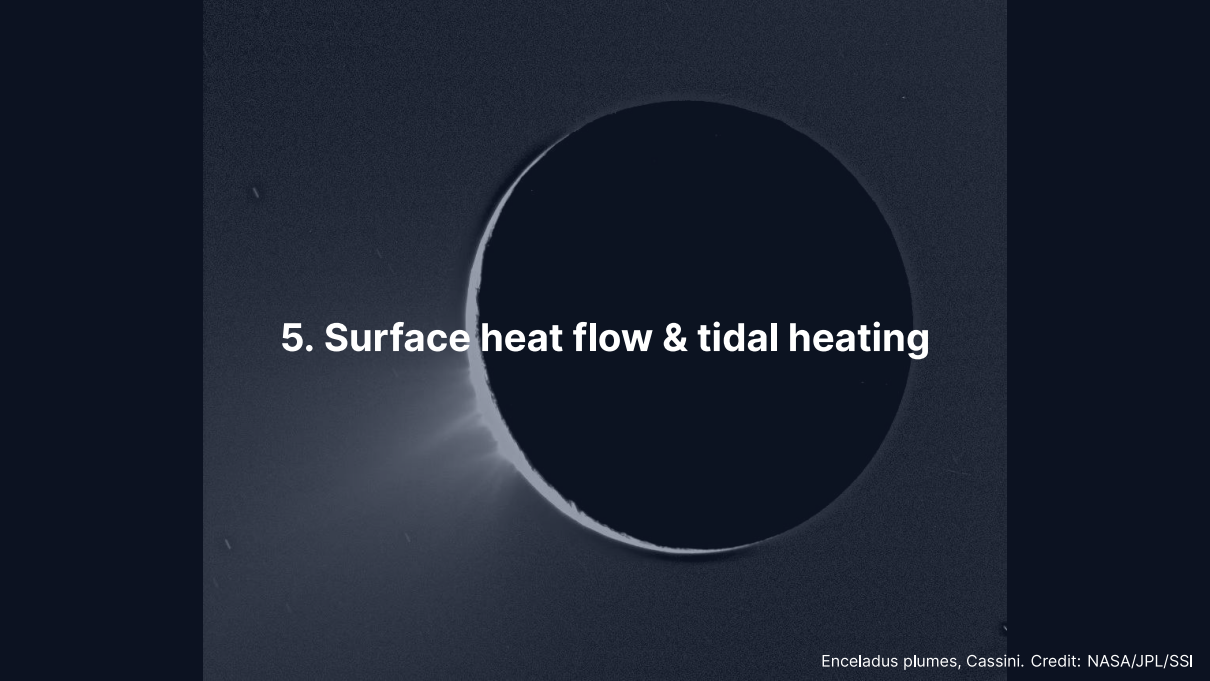
Plate boundaries and selected major hotspots. Credit: USGS, public domain.

Reading the global hotspot map



- Divergent, convergent, and transform boundaries partition the surface into plates.
- Hotspots (dots) sit both on ridges (Iceland, Azores) and deep inside plates (Hawaii, Yellowstone).
- Intraplate hotspots are the surface expression of plumes rooted near the core-mantle boundary, fixed while plates drift over them.

Hotspot chains are the mantle's own reference frame: they record plate motion over the deep mantle.



5. Surface heat flow & tidal heating

Core cooling and the dynamo

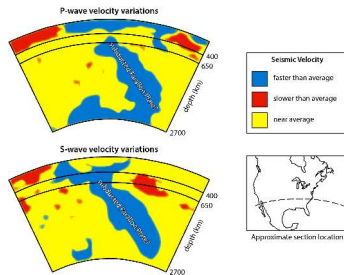
The mantle acts as a thermostat on the core.

- Heat flux from core to mantle sets the rate of core cooling
- A vigorously convecting mantle extracts heat efficiently, enabling core convection
- Core cooling drives the inner core's gradual crystallisation
- Inner core growth releases latent heat + light elements → compositional buoyancy
- This compositional convection is the **dominant** driver of today's geodynamo (Lecture 4)

Earth's magnetic field ultimately exists because mantle convection efficiently cools the core; stagnant-lid Venus likely lacks this.

Whole-mantle vs. layered convection

The debate: does Earth's mantle convect as one layer or two?



Credit: Wikimedia (USGS/Caltech), CC BY-SA

- Seismic tomography images the Farallon slab descending to the core–mantle boundary
- Plumes rise from the hot bottom boundary layer through the full mantle
- Strong evidence for **whole-mantle convection**
- A partial barrier at 660 km slows, but does not stop, slabs; the mantle convects as one layer

Source: Grand (2002); Ritsema et al. (2011)

Plumes vs. plates: two modes of upwelling

- **Mid-ocean ridges:** passive upwelling where plates pull apart
- Temperatures close to the ambient mantle
- **Mantle plumes:** active upwelling from the deep mantle
- ~200–300 K hotter than the ambient mantle (from olivine thermobarometry)
- Plumes deliver primordial (^3He -rich) material from the lower mantle
- Large Igneous Provinces (LIPs) may be plume-heads: Deccan Traps, Siberian Traps, Ontong-Java Plateau
- LIPs correlate with mass extinction events in geological history

Source: Courtillot & Renne (2003), *Comptes Rendus Geoscience*

Surface heat flow

Body	Heat flux (mW m^{-2})	Total (TW)	Source
Earth	~90	~47	Radiogenic + primordial
Moon	10–15	~0.3	Primordial (cooled)
Mars	15–25	~3–5	Radiogenic + primordial
Io	2000–2500	~105	Tidal dissipation

Source: Davies & Davies (2010); Davies et al. (2024); InSight + gravity/topography estimates

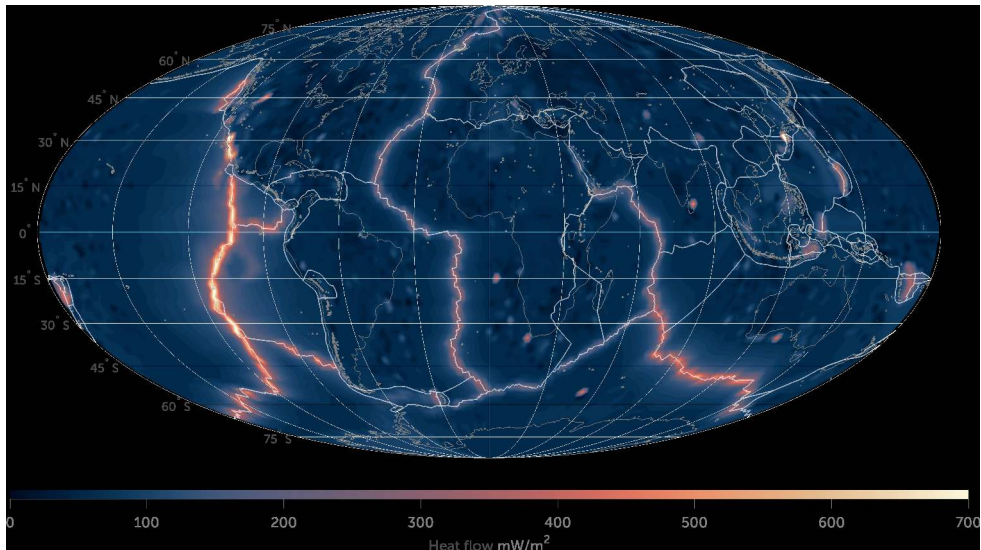
Io's heat flux is 20–30× higher than Earth's per unit area, and its total power (105 ± 12 TW) is roughly **twice** Earth's 47 TW, from a body with only 1.5% of Earth's mass.

Spatial pattern of Earth's heat flow

Earth's 47 TW is not uniformly distributed:

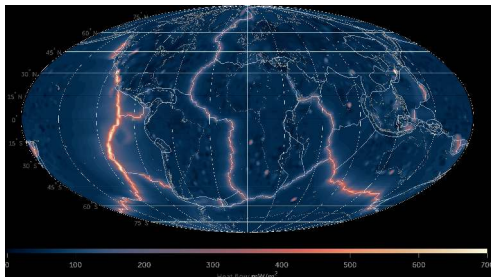
- **Oceanic regions** (~60% of area): 70% of total heat loss
 - Thin lithosphere, new crust at ridges
 - Flux decreases with crustal age (Stein & Stein 1992 cooling model)
- **Continental regions** (~40% of area): 30% of total heat loss
 - Thick lithosphere insulates the mantle
 - BUT concentration of U, Th, K in the crust adds local contribution
 - ~40% of the continental surface flux is crustal-radiogenic

Source: Jaupart & Mareschal (2011); Davies & Davies (2010)



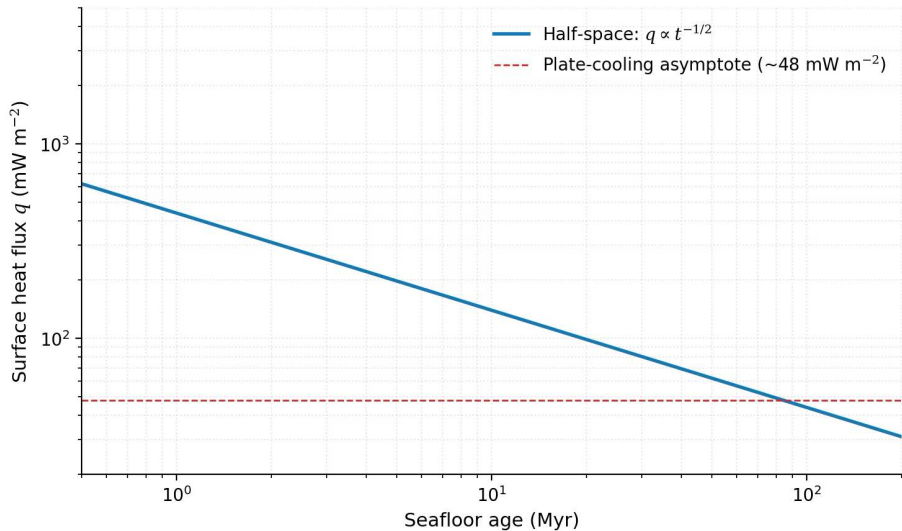
Earth's surface heat flow. Credit: Fabio Crameri (Wikimedia, CC BY-SA 4.0); data from Lucazeau (2019).

Reading the global heat-flow map



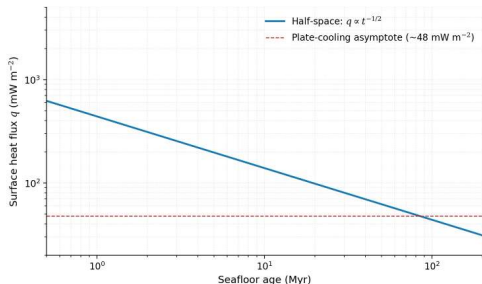
- Mid-ocean ridges blaze as the hottest features: young, thin lithosphere above upwelling mantle.
- Heat flow falls with seafloor age away from each ridge, exactly as the half-space model predicts.
- Old continental shields are the coldest surfaces: thick, insulating, ancient lithosphere.
- Integrating the map gives the 47 TW total; the compilation behind it holds ~ 70,000 borehole measurements.

The heat-flow map is plate tectonics made visible in a single physical quantity.



Seafloor heat flux against crustal age. Course figure; plate-cooling asymptote after Stein & Stein (1992).

Reading the heat flux vs age curve



- The blue curve is the half-space prediction $q \propto t^{-1/2}$: the same erf-solution as before, differentiated at the surface.
- At old ages the data level off at the plate-cooling asymptote ($\sim 48 \text{ mW m}^{-2}$): the lithosphere reaches an equilibrium thickness.
- Over young crust, measured values fall *below* the curve because hydrothermal circulation carries part of the heat.

One curve connects the heat equation to ship-borne heat-flow surveys, and its deviations are physics too.

Tidal heating: how it works

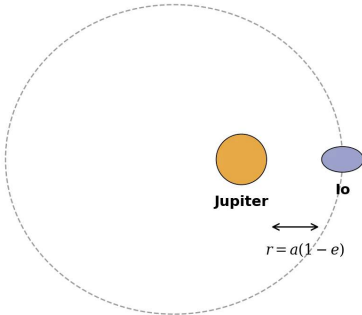
- Circular orbit + tidal lock → static bulge, **no heating**
- **Eccentric orbit:** distance varies each orbit
 - Closer: tidal force stronger, moon moves faster
 - Farther: tidal force weaker, moon moves slower
- Continuously varying deformation → **flexing dissipates heat**

Heating depends on:

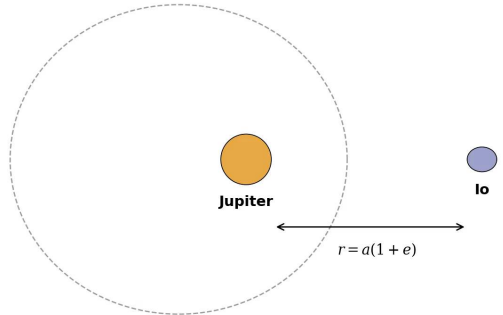
- Eccentricity e (zero $e \rightarrow$ zero heating)
- Distance a : heating $\propto a^{-15/2}$ at fixed e (the explicit a^{-6} times $n \propto a^{-3/2}$)
- Internal dissipation (tidal quality factor Q)

Cyclic flexing dissipates orbital energy as heat: $\dot{E}_{\text{tidal}} \propto e^2/Q$

Periapsis: large tidal bulge

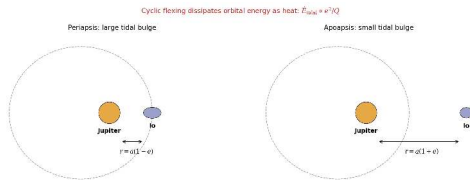


Apoapsis: small tidal bulge



Tidal flexing on an eccentric orbit: the origin of tidal heating. Course schematic.

Reading the tidal-flexing schematic



- **Left, periapsis** ($r = a(1 - e)$): the moon is closest and its tidal bulge is largest.
- **Right, apoapsis** ($r = a(1 + e)$): the bulge is smallest.
- Each orbit, the body is squeezed and stretched between the two states; internal friction turns that cyclic work into heat at a rate $\propto e^2/Q$.

Eccentricity is the on-off switch of tidal heating: no e , no flexing, no heat.

Secular cooling of planets

Planets are thermally non-stationary: they started hot and are gradually cooling.

- Stored heat $\sim E_{\text{acc}} + E_{\text{diff}} + \int L_{\text{rad}}(t) dt$
- Lost heat $\sim \int L_{\text{surface}}(t) dt$
- The difference is the present-day thermal state
- Earth: still cooling at $\sim 50\text{--}100$ K/Gyr
- Smaller bodies cool faster (lower thermal inertia, thinner conductive boundary layer)

All terrestrial planets are in net secular cooling; Earth's is slowed by convective resurfacing but not stopped.

Tidal dissipation rate

For a synchronously rotating body on an eccentric orbit, the tidal heating power is (to leading order in e):

$$\dot{E}_{\text{tidal}} = \frac{21}{2} \frac{k_2}{Q} \frac{GM_*^2 R^5 n e^2}{a^6}$$

where:

- k_2 = Love number (dimensionless, ~ 0.3 for rocky bodies)
- Q = tidal quality factor (lower = more dissipative)
- M_* = mass of the primary (Jupiter, Saturn, etc.)
- R = radius of the moon
- n = mean motion ($= \sqrt{GM_*/a^3}$); a , e = semi-major axis and eccentricity

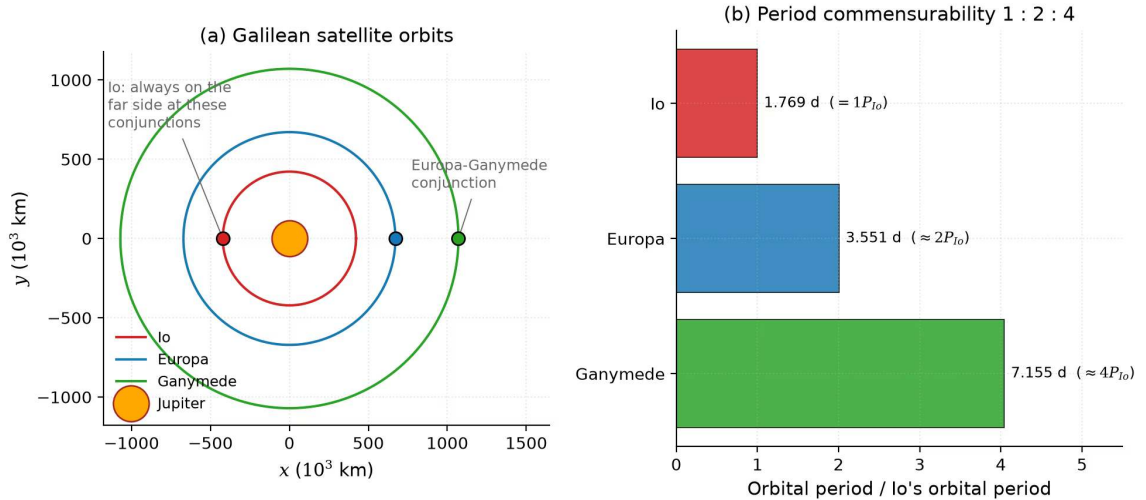
Note the steep R^5/a^6 dependence: heating is dramatic for close-in moons.

The Laplace resonance: sustaining eccentricity

Io's orbit would circularise in ~ 10 Myr if left alone. Why is its eccentricity maintained?

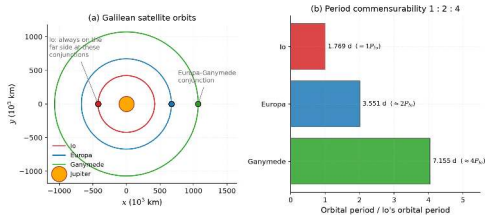
- Io–Europa–Ganymede are locked in a 4:2:1 **mean-motion resonance**
- Every time Io laps Europa, the two meet at the same orbital phase
- Periodic gravitational kicks continually **pump** Io's eccentricity
- Without this, tidal dissipation would damp e and shut off the heating
- The resonance was first identified by Laplace in 1805

The three-body resonance is what makes Io's tidal heating indefinite, not transient.



The Laplace resonance of Io, Europa, and Ganymede. Course figure.

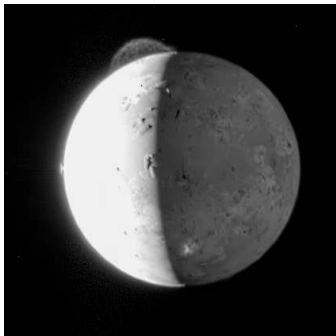
Reading the Laplace resonance figure



- **Panel (a):** the three orbits to scale at a Europa-Ganymede conjunction; the phase lock puts Io on the *opposite* side, so a triple conjunction never occurs.
- **Panel (b):** the periods stand in the ratio 1 : 2 : 4, and the conjunction pattern repeats every Ganymede orbit.
- The repeating kicks force Io's eccentricity to $e \approx 0.004$: small, but enough.

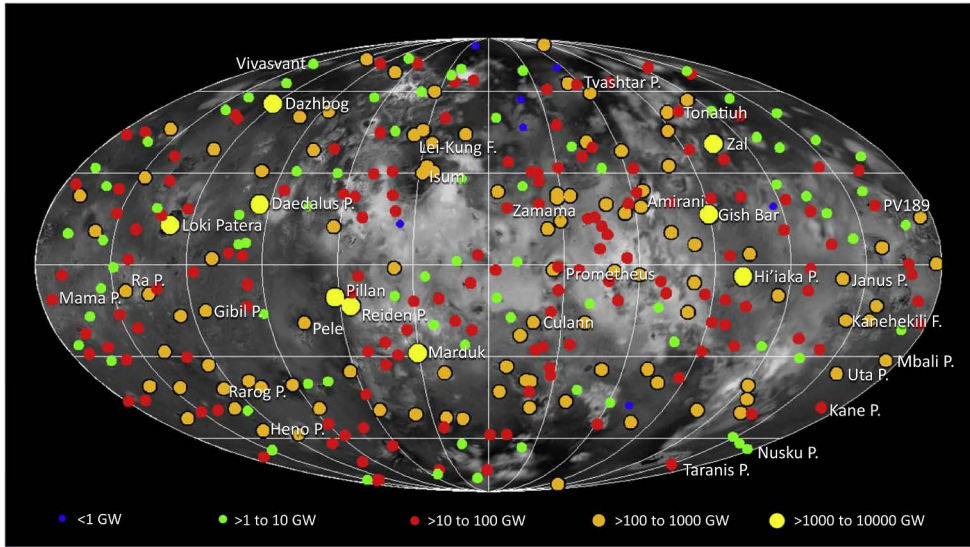
An orbital clockwork three moons wide is what keeps a single moon volcanically alive.

Io: the most volcanic body



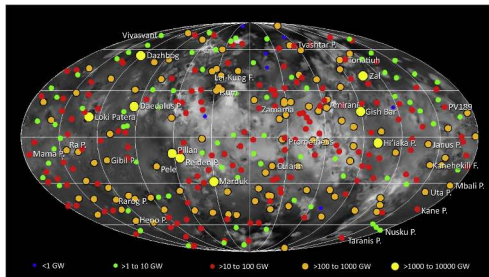
Credit: NASA/JHUAPL/SwRI (2007)

- $\sim 10^{14}$ W of tidal heat (2× Earth's total!)
- Predicted by Peale, Cassen & Reynolds (1979)
- Confirmed days later by Voyager 1
- Powered by the **Laplace resonance**
- 343 catalogued active volcanic centres (Davies et al. 2024)
- Entire surface resurfaced on $\sim$ Myr timescale



Io's 343 catalogued volcanic hot spots from 30 years of monitoring. Reproduced from Davies et al. (2024), Fig. 1.

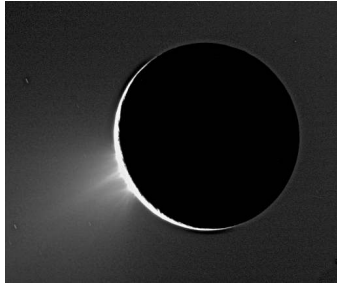
Reading Io's hot-spot map



- Marker size and colour scale with each hot spot's power, from blue to yellow; Loki Patera tops the list at 6000–12,000 GW.
- The catalogued spots emit 58 ± 1 TW, about 55% of Io's 105 ± 12 TW total; the rest leaves through buried and subsurface volcanism.
- Hot spots occur at all latitudes at the same areal density, but polar volcanoes are weaker: they emit half the energy of lower-latitude ones, favouring shallow over deep-mantle tidal heating.

A body 1.5% of Earth's mass out-radiates Earth twice over, entirely on orbital energy.

Enceladus: a tiny moon with a big secret



Credit: NASA/JPL/SSI (Cassini)

- Radius only 252 km
- Water ice jets from “tiger stripe” fractures
- Thermal power: ~16 GW (tidal dissipation)
- Global subsurface ocean beneath ~15–25 km ice
- Plumes carry salts, silica, organics, and H_2 : hydrothermal activity on the ocean floor

Tidal heating can sustain liquid-water oceans far outside the habitable zone.

Where in Enceladus is tidal heat dissipated?

Two competing scenarios explain the observed ~16 GW of thermal power:

Rocky core dissipation:

- Tidal flexing of porous silicate core
- Heat conducted up through the ocean
- Matches hydrothermal signatures (silica nanoparticles at ~90°C)
- Difficulty: plume power exceeds simple core-heating estimates

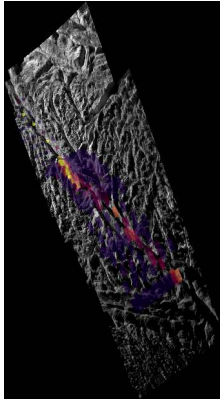
Ice-shell dissipation:

- Viscous dissipation concentrated in the warm ice shell base
- Positive feedback: softer ice → more dissipation
- Can reach 10–20 GW with realistic rheology
- Concentrates heat near the south pole tiger stripes

Likely: both contribute, with the ice shell dominating the plume heat flux.

Source: Čadež et al. (2016); Hemingway & Mittal (2019)

Where the heat leaves Enceladus



- A high-resolution Cassini CIRS thermal scan (colour) along one tiger-stripe fracture, draped on the visible-light mosaic.
- The warm emission hugs the fracture and dies off within a few km to either side: the heat leaves through the cracks, not the plains.
- Integrated over the whole south polar terrain: 15.8 ± 3.1 GW, far beyond any radiogenic budget for a 252 km moon.

On Enceladus, tidal heat is not diffuse: it is plumbed to the surface along four fractures.

Enceladus plumes: a habitability laboratory

Cassini's multiple plume flythroughs revealed:

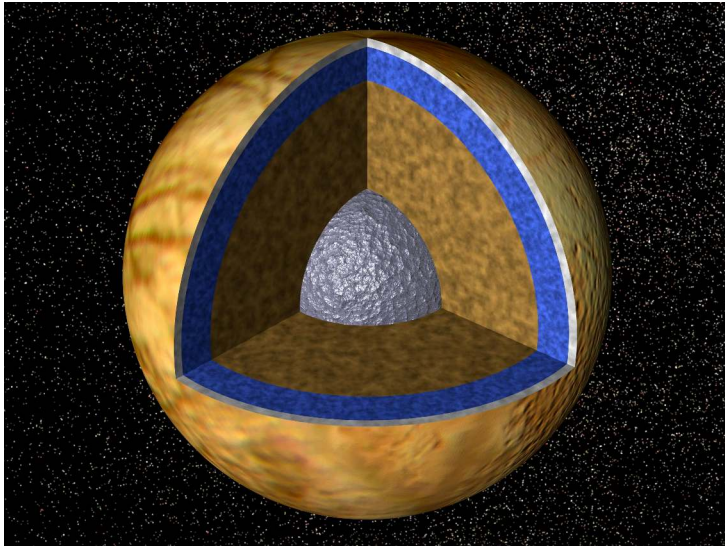
- **Salts:** NaCl, NaHCO₃: direct samples of the subsurface ocean
- **Silica nanoparticles:** imply ocean-floor temperature >90°C (hydrothermal)
- **Molecular hydrogen** (H₂): produced by water-rock serpentinisation
- **Organic molecules:** CO₂, CH₄, NH₃, HCN, complex species up to C₆
- Phosphate detected in 2023 (Postberg et al.), last major missing bioessential element

Enceladus' ocean has all the ingredients (liquid water, energy, organics, phosphorus) for chemoautotrophic life, if it exists there.

Ocean worlds: Europa & Titan

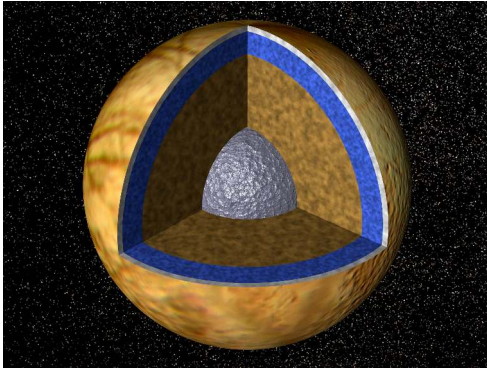
- **Europa:** Global ocean ~100 km deep, ice shell ~15–25 km
Maintained by Laplace resonance with Io and Ganymede
Europa Clipper mission (launched 2024) will investigate habitability
- **Titan:** Subsurface water–ammonia ocean
Evidence from rotational state (Cassini)
Dragonfly mission (launch ~2028, arrival ~2034) will explore surface chemistry

These ocean worlds are among the most promising targets for extraterrestrial life: they have liquid water, chemical energy, and organic building blocks.



Europa's inferred interior: iron core, rocky mantle, global ocean, ice shell. Credit: NASA/JPL.

Reading Europa's interior cutaway



- From the centre outward: an iron-nickel core, a silicate mantle, a global liquid-water ocean ~100 km deep, and an ice shell of ~15–25 km.
- Tidal dissipation in the shell and silicate interior, forced by the Laplace resonance, keeps the ocean from freezing.
- Magnetic induction and gravity measurements by Galileo are the evidence; Europa Clipper will measure the shell directly.

More liquid water than all of Earth's oceans, kept warm by the same resonance that melts Io.

A grayscale image of the planet Io, showing its heavily cratered surface. A bright, elongated plume of material is visible on the left side of the planet, extending into the dark space above it. The plume has a distinct, bright, and somewhat irregular shape, contrasting with the darker, cratered surface of the planet.

6. Recent advances & outlook

Recent advances

- **InSight mission** (2018–2022): first seismology on Mars
Seismic core radius ~1830 km; reanalyses place a ~1675 km metal core beneath a molten silicate layer
Heat flow probe encountered unexpected soil properties
- **Tidal heating models:** Updated rheological models suggest dissipation in Enceladus may be concentrated in the ice shell
- **Radiogenic budget:** Updated meteorite analyses refine early solar system heating

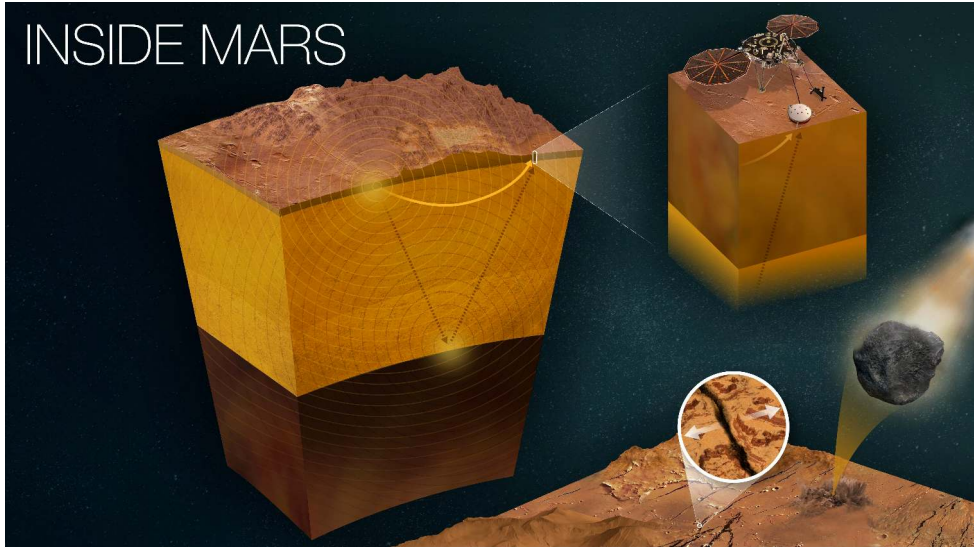
InSight: Mars heat flow and interior

InSight's seismometer (SEIS) and heat flow probe (HP³) transformed our view of Mars.

- Over 1300 marsquakes detected during the mission
- Seismic core radius: 1830 ± 40 km, larger than pre-mission estimates
- Reanalyses (2023): a ~1675 km metal core beneath a ~150 km molten silicate layer
- Crustal thickness: 24–72 km (two-layer model)
- Heat flow probe stuck at 40 cm depth in unexpectedly cohesive regolith
- Upper limit on near-surface heat flux: $\sim 22 \text{ mW m}^{-2}$

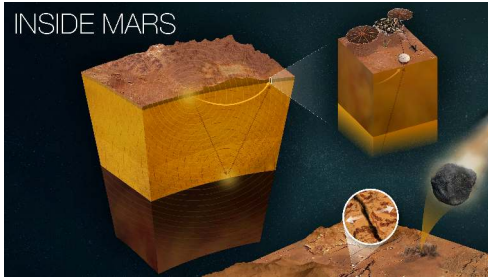
Source: Stähler et al. (2021); Khan et al. (2021, 2023); Samuel et al. (2023); Spohn et al. (2022)

INSIDE MARS



How InSight's marsquake travel times constrain Mars's interior. Credit: NASA/JPL-Caltech.

Reading the InSight cutaway



- Seismic waves from marsquakes and impacts travel along depth-dependent paths; their arrival times encode the interior structure.
- SEIS (the seismometer) fixed the crustal thickness, the mantle structure, and the core radius; HP³ probed the regolith's thermal properties.
- The 2023 reanalyses add a ~150 km molten silicate layer atop a ~1675 km metal core.

One seismometer turned Mars from a one-dimensional guess into a measured interior.

Europa Clipper: targeting the ice shell

NASA Europa Clipper launched October 2024, arrives at Jupiter in 2030.

- Will make ~50 close flybys of Europa over ~4 years
- **Ice Penetrating Radar (REASON)**: maps ice shell thickness and near-surface layers
- **Magnetometer**: constrains ocean salinity from induced response
- **MASPEX + SUDA**: samples the composition of possible plume material
- Thermal mapping will locate regions of thinner ice and recent activity
- Will directly test whether tidal dissipation is concentrated in the shell or in the rocky core

Source: Howell & Pappalardo (2020), *Nature Communications*

Modelling Mars' thermal evolution

Mars' thermal history explains key observations:

- Early intense volcanism (Tharsis) → hot magma ocean + dynamo (4.1 Ga)
- Dynamo shutdown ~4.1 Ga → core cooling slowed, crustal magnetisation frozen
- Olympus Mons still eruptive until ~100–200 Ma
- Present-day heat flow ~15–25 mW/m² (InSight constraints)
- Partially molten upper core retains heat; slow mantle convection prolongs cooling

Mars' thermal history ties directly to its atmospheric evolution: the loss of the dynamo caused the loss of the field, and eventually the atmosphere.

Io in the decade of Juno

Juno's extended mission (2021–) has delivered the closest-ever Io observations:

- Global heat flow map: polar regions are hotter than expected, contradicting simple tidal models
- Thermal emission imaging from JIRAM instrument
- Gravity data consistent with a global magma ocean just beneath the crust (Park et al. 2024)
- New implications for the structure of Io's interior: is the “mantle” largely molten?
- Future: IVO (Io Volcano Observer) concept proposal for dedicated Io mission

Source: Park et al. (2024); Keszthelyi et al. (2022)

JUICE: The Jupiter Icy Moons Explorer

ESA's JUICE launched in April 2023, arrives Jupiter 2031.

- Multiple flybys of Europa, Callisto, and Ganymede
- Orbit insertion at **Ganymede** in 2034: first orbiter of a moon other than our own
- Will determine the thickness of Ganymede's ice shell and ocean via laser altimetry and induced magnetic field
- Direct measurement of tidal heating through tidal Love number k_2
- Complementary to Europa Clipper: two flagship ocean world missions in the 2030s

Source: Grasset et al. (2013), *Planet. Space Sci.*

Are Ocean Worlds Habitable?

- Require **three things**: liquid water, energy, chemical nutrients
- **Liquid water**: tidal heating + ice shell insulation
- **Energy**: chemical disequilibria (serpentinisation produces H_2)
- **Nutrients**: rock–water reactions deliver P, S, trace metals
- **Open questions**: redox state, temperature range, ocean pH, mixing timescales
- Ganymede, Europa, Enceladus, Titan, Triton, Pluto and maybe Ceres all have possible subsurface oceans

Source: Nimmo & Pappalardo (2016), *J. Geophys. Res. Planets*

Summary

1. Four heat sources: accretional, gravitational differentiation, radioactive decay, tidal heating
2. Conduction is diffusive ($\tau \sim L^2/\kappa$), too slow for planet-sized bodies
3. Convection dominates in large bodies: $Ra \gg Ra_c$
4. Earth: mobile lid (plate tectonics); other rocky bodies: stagnant lid
5. Tidal heating powers Io's volcanism and maintains subsurface oceans on Enceladus, Europa, and Titan
6. Earth's total heat loss: 47 TW (40% radiogenic, 60% primordial)

Looking ahead to Lecture 4

This lecture followed the **heat**: where it comes from and how it escapes.

Lecture 4 follows the **chemistry** the heat enables:

- How a molten planet separates into a metallic core and a silicate mantle.
- How siderophile and lithophile elements sort themselves between the two.
- What the Hf–W chronometer says about how *fast* core formation happened.
- How a convecting metallic core generates the magnetic field that shields an atmosphere.

The heat sources assembled today set the pace for all of it: differentiation needs melting, and the dynamo needs a cooling core.

Questions?

Next: Lecture 4, Chemical differentiation & magnetospheres

Thu 17 Sep, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 18 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

<https://ips.formingworlds.space>